

DSA Patterns — Interview Priority Guide

23 pattern categories reordered by importance for coding interviews (FAANG-style)

Tier 1 — Must-master (asked constantly, foundational)

01. Arrays & Hashing

- Frequency / Hash Map
- Subarrays / Subsequences
- In-place Array Techniques
- Matrix Manipulation
- Coordinate Compression

02. Two Pointers & Sliding Window

- Two Pointers
- Fast & Slow Pointers
- Fixed Window
- Variable Window
- Frequency-based Window

06. Stack, Queue & Deque

- Stack
- Monotonic Stack
- Queue
- Monotonic Queue
- Deque
- Expression / Parentheses

11. Binary Tree

- DFS
- BFS / Level Order
- Tree Traversals
- Path Problems
- LCA
- Tree Construction
- Tree DP

14. Graphs

- Graph Representation
- DFS / BFS
- Grid DFS / BFS
- Multi-Source BFS
- Connected Components
- Cycle Detection
- Bipartite Graph
- Topological Sort
- DSU / Union Find

16. Dynamic Programming

- 1D DP
- 2D / Grid DP
- Knapsack
- Subset / Partition DP
- LIS / LCS
- String DP
- Interval DP
- Tree DP
- Bitmask DP
- Digit DP

03. Binary Search

- Classic Binary Search
- Lower / Upper Bound
- First / Last Occurrence
- Rotated Array
- Binary Search on Answer

07. Linked List

- Reversal
- Partial Reversal
- Fast / Slow
- Cycle Detection
- Merge
- Intersection
- Random Pointer

Tier 2 — Very common, high ROI

10. Recursion & Backtracking

- Basic Recursion
- Subsets
- Permutations
- Combinations
- Combination Sum
- Constraint Problems
- Grid Backtracking

15. Greedy

- Greedy + Sorting
- Greedy + Heap
- Interval Greedy
- Scheduling
- Reachability
- Exchange / Choice-based Greedy

09. Heap / Priority Queue

- Min / Max Heap
- Top K
- Two Heaps
- K-Way Merge
- Heap + Greedy

05. Prefix Sum & Difference Array

- 1D Prefix Sum
- 2D Prefix Sum
- Difference Array
- Range Queries

18. Strings & String Algorithms

- Palindrome
- Anagram
- Frequency
- String Matching
- KMP
- Z Algorithm
- Rolling Hash

08. Intervals & Sweep Line

- Merge Intervals
- Insert Interval
- Overlapping Intervals
- Interval Scheduling
- Sweep Line

04. Sorting & Partitioning

- Merge Sort
- Quick Sort
- Quickselect
- Cyclic Sort
- Counting / Radix Sort
- 2-Way Partition
- 3-Way Partition / DNF
- Stable Partition
- Inversion Counting

12. Binary Search Tree (BST)

- Search / Insert / Delete
- Validation
- Kth Smallest / Largest
- LCA
- Predecessor / Successor

Tier 3 — Common at mid/senior level or specific companies

17. Bit Manipulation & Bitmasking

- XOR
- Bit Operations
- Bit Tricks
- Bit Masks
- Bitmask Enumeration

13. Trie

- Prefix Search
- Word Search
- Autocomplete
- Trie + Backtracking
- Bitwise Trie

19. Math & Number Theory

- GCD / LCM
- Prime / Sieve
- Modular Arithmetic
- Combinatorics
- Fast Power
- Probability
- Geometry

23. Miscellaneous / Special Techniques

- Simulation
- Design / Data Structure Implementation
- Game Theory
- Parsing / Implementation
- State Machines
- Special Mathematical Tricks
- Problems that don't fit elsewhere

Tier 4 — Senior/staff, infra-heavy, or specialized roles

21. Advanced Graph Algorithms

- Dijkstra
- Bellman-Ford
- Floyd-Warshall
- 0-1 BFS
- MST
- Kruskal
- Prim
- Binary Lifting

20. Advanced Data Structures

- Fenwick Tree / BIT
- Segment Tree
- Lazy Propagation
- Sparse Table
- Ordered Set / Multiset

Tier 5 — Rare in interviews, mostly contest/niche

22. Advanced Techniques

- Divide & Conquer
- Meet in the Middle
- Sweep Line Extensions
- Randomized Algorithms
- Reservoir Sampling
- Contribution Technique
- Offline Queries

Practical guidance: Tier 1 + Tier 2 alone cover the overwhelming majority of standard tech interviews (including most FAANG phone screens and onsites). Tier 3 shows up more at companies known for asking trickier questions or for senior roles. Tiers 4-5 are mostly relevant for infra/systems-heavy roles, competitive programming, or a handful of companies (e.g., some quant/HFT firms) — don't burn early prep time there unless you know your target company tests it.